

Durability of glass fiber-reinforced polymer bars conditioned in moist seawater-contaminated concrete under sustained load

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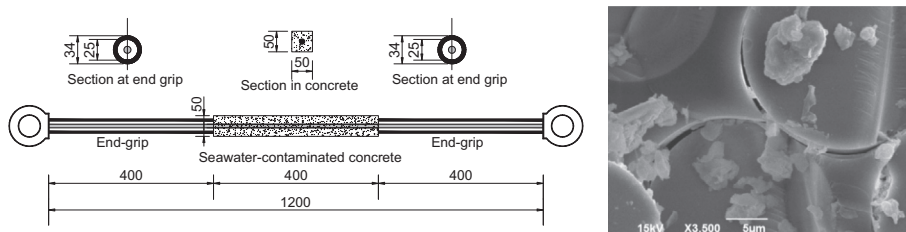
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HIGHLIGHTS

- Durability of GFRP conditions in concrete environment under load was investigated.
- GFRP bars experienced creep rupture were analyzed for microstructure changes.
- New durability design models for GFRP bars in severe environments were developed.

GRAPHICAL ABSTRACT



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ABSTRACT

The combined effect of sustained loading and moist seawater-contaminated concrete environment on the durability performance of two types of glass fiber-reinforced polymer (GFRP) bars was investigated. A sustained load of 25% of the ultimate tensile strength was applied to concrete-encased GFRP bars immersed in tap water for up to 15 months at temperatures of 20, 40, and 60 °C. None of the loaded specimens conditioned at 20 °C were creep-ruptured during conditioning. In contrast, many bars were creep-ruptured during conditioning at the higher temperatures of 40 and 60 °C. At a conditioning temperature of 20 °C, Type I and II GFRP bars retained 84 and 70% of their original tensile strength after 15 months of conditioning under a sustained load. At the higher temperature of 60 °C, trivial tensile strength retentions of 15 and 12% were recorded for Type I and Type II GFRP bars, respectively, after 15 months of conditioning under a sustained load. A comparative analysis between results of loaded and unloaded specimens demonstrated the detrimental effect of the presence of a sustained load during conditioning on the tensile strength retention of GFRP bars. The detrimental effect of the sustained load was intensified at the higher temperatures. The accelerated aging test data along with Arrhenius concept were employed to develop a durability design model that can predict the tensile strength retention of both types of GFRP bars in moist seawater-contaminated concrete.

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1. Introduction

The use of glass fiber-reinforced polymer (GFRP) bars as a replacement to the traditional steel reinforcement would eliminate corrosion problems and prolong service life of reinforced concrete structures [1–3]. Even with their superior performance and

relatively competitive cost, the use of GFRP has been hindered in regions of high temperature and high relative humidity due to concerns about their durability performance under sustained loading in such harsh environmental conditions.

GFRP bars conditioned in simulated pore solution (pH 13.5) under a sustained load of 10% of the ultimate tensile stress (UTS) lost 70% of their initial tensile strength after 9 months of conditioning [4]. Increasing the sustained load to 15 and 25% of UTS resulted in a creep-rupture of GFRP specimens within 100 and 20 days,

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respectively [4]. Nkurunziza et al. [5] assessed the creep behavior of GFRP reinforcing bars enclosed by either an alkaline solution (pH 12.8) or de-ionized water (pH 7.0) at two percentages of sustained tensile stresses of 25 and 38% UTS for 417 days. While the tensile strength of the GFRP bars after conditioning in de-ionized water was unaffected, conditioning in alkaline solution under sustained loading of 25 and 38% of UTS caused tensile strength reductions of 12 and 32%, respectively. In either loading case, conditioned specimens did not experience any creep rupture [5]. Debaiky et al. [6] studied the combined effect of sustained loading and harsh environment on the performance of GFRP bars immersed in water and alkaline solution. The reduction in tensile properties of sand-coated GFRP bars conditioned at elevated temperatures in alkaline and water environments under sustained load were in the range of 7–13%.

The presence of sustained load during conditioning in concrete or mortar-wrapped GFRP bars could intensify degradation of GFRP bars [7,8]. Increasing the stress levels caused a shift in the degradation mechanism of GFRP from being influenced by the diffusion rate of alkaline solutions to being controlled by solution transport through resin cracks at higher stress levels [9]. Davalos et al. [10] investigated the performance of GFRP bars exposed to concrete environment with and without displacement-controlled load for up to 210 days at temperatures between 20 and 60 °C. None of the GFRP bars experienced creep-rupture. Loaded and unloaded specimens exhibited similar tensile strength retentions with conditioning at 60 °C being most critical [10].

Long-term performance of concrete specimens reinforced with GFRP bars exposed to different environmental conditions has also been investigated [11–14]. Experimental results indicated that the strain in GFRP bars was highest in RC beams subjected to seawater at 40 °C for 300 days with cyclic wetting and drying [11]. Fergani et al. [12] reported that the tensile properties of GFRP bars embedded in concrete was critically affected by exposure to elevated temperature of 60 °C rather than sustained stress levels (up to 5000 $\mu\epsilon$). Microstructure investigations of these GFRP bars showed a degradation of the resin matrix, formation of microcracks due to a hydrolysis reaction [13]. Additionally, a stress level corresponding to in-service conditions complied with limits set by current guidelines, while higher loads and wet conditioning resulted in excessive deflections and crack widths that could not be predicted by codified models [14]. In other work, Laoubi et al. [15] investigated the behavior of GFRP bars imbedded in concrete beams and subjected to sustained load and 100, 200, and 360 freeze-thaw cycles. The authors concluded that the coupled action of sustained loading and harsh conditioning had no significant effect on the performance of the GFRP reinforcement or the concrete beams. On the contrary, it was found that the limits set by ACI 440.1R [16] were conservative. Robert and Benmokrane [17] reported data on GFRP specimens that were preloaded up to 80% of the UTS prior to conditioning at temperatures between 23 and 50 °C for 240 days. Results highlighted a progressive hydrolysis reaction and up to 11% tensile strength reduction [17]. Robert et al. [1], Almusallam and Al-Salloum [18], and Trejo et al. [19], concluded that GFRP bars imbedded in concrete exposed to harsh environments or alkaline solutions did not properly simulate those left in natural conditions.

Despite the significant amount of research conducted in this area, the tensile strength reduction of GFRP bars caused by different environmental exposures and levels of sustained loading has witnessed wide variations. While several studies have been conducted on durability of GFRP bars in concrete environment, the long-term performance of GFRP bars in seawater-contaminated concrete under sustained load has received little attention. Seawalls, marine docks and waterfront/offshore concrete structures are continuously exposed to seawater and/or seawater splash.

Although seawater is not used in steel-reinforced concrete structures because of the corrosion problem, there might be a potential to use seawater in concrete structures reinforced with non-metallic corrosion-resistant composite reinforcement such as GFRP bars. The use of seawater in steel-free concrete structures would preserve natural resources and promote sustainability, particularly in desert areas where water is scarce and expensive. More research is, therefore, needed to examine the behavior of GFRP in such environments and facilitate the development of a durability design model of GFRP reinforcing bars in seawater-contaminated concrete structures.

This paper aims to provide an improved understanding into the durability performance of two different types of commercially-produced GFRP bars conditioned in moist seawater-contaminated concrete under a sustained load. The ACI 440.1R [16] limits the tensile stress in GFRP under service conditions to 14% of UTS, considering the strength reduction factor for environmental exposure. Specimens of the present study were conditioned at a stress level of 25% of UTS to investigate the possibility of increasing the allowable limit set by the ACI 440.1R [16]. Tensile strength retentions of GFRP bars conditioned under a sustained load were measured. Bars experienced creep rupture were analyzed for microstructure changes while highlighting the influence of hydrolysis reaction on mechanical and durability properties. New durability design models were developed based on Arrhenius concept to predict the long-term tensile strength retention of GFRP bars in moist seawater-contaminated concrete.

2. Material details

2.1. GFRP bar specimens

In this study, two types of ribbed GFRP bars made of high strength continuous glass fibers impregnated in epoxy resin were used as shown in Fig. 1. Physical and mechanical properties of Type I and II GFRP bars and corresponding ASTM standard procedures are summarized in Table 1. Both types of GFRP bars had similar mass fraction of glass fibers. Such fiber content included fibers and fillers, as reported in ASTM D3171 [20]. As shown in Table 1, the voids content in Type II GFRP was higher than that of Type I. Also, the as-received ultimate tensile strength was higher in Type II than in Type I GFRP.

2.2. Surrounding concrete

GFRP test specimens were cut into 1200 mm lengths. The test region had a length of 400 mm, given the minimum length set by the ACI 440.3R [21] as 40 times the bar diameter. The middle third (test region) was encased by 50 × 50 mm seawater-contaminated concrete to simulate concrete exposed to seawater in field conditions. The ACI 318 building code [22] requires a minimum concrete cover of 40 mm for steel-reinforced concrete structures, with a steel bar diameter ≤ 16 mm, exposed to earth or weather. The use of GFRP bars would eliminate corrosion problems; hence, a thinner concrete cover maybe adequate for GFRP-reinforced concrete structures. Further research is certainly needed to investigate the effect of varying the thickness of the concrete cover on the durability of GFRP bars in severe environments. The 400 mm-long end regions of the bars were used as grips for applying the sustained load. Table 2 presents the mixture proportions of the surrounding concrete. The main binding agent of concrete was ASTM Type I ordinary



Fig. 1. GFRP bars used in the current study.

Table 1

Physical and mechanical properties of Type I and II GFRP bars.

Property	Unit	Standardized Testing Procedure	GFRP Type	
			Type I	Type II
Diameter (inner/outer)	mm	ASTM D7205 [27]	7.2/8.0	8.0/9.0
Cross-sectional area	mm ²	ASTM D7205 [27]	45.0	54.0
Mass glass fraction	%	ASTM D3171 [20], ASTM E1868 [40]	78.0	75.0
Void content, by volume	%	ASTM D3171 [20], ASTM D2734 [41]	0.10	0.23
Glass transition temperature	°C	ASTM E1356 [28]	106.0	125.0
Ultimate tensile strength	MPa	ASTM D7205 [27]	816	1321

Table 2

Surrounding concrete mix proportions.

Unit weight (kg/m ³)				
Cement	Fine aggregate	Coarse aggregate	Seawater	w/c
400	580	1160	200	0.5

Table 3

Physical and mechanical properties of surrounding concrete.

Property	Unit	Standardized Testing Procedure	Value
Compressive strength	MPa	ASTM C39 [42]	40.0
Tensile splitting strength	MPa	ASTM C496 [43]	3.2
Ultra-pulse velocity	m/s	ASTM C597 [44]	5405.0
Electrical conductivity	kΩ.cm	ASTM C1760 [23]	8.0
RCPT	Coulombs	ASTM C1202 [24]	2427.0
pH	n/a	ASTM E70 [45]	12.4

Portland cement. Natural crushed stone with a nominal maximum size of 10 mm served as coarse aggregates, while a blend of equal proportions of dune sand and natural crushed stone were used as fine aggregates. Seawater obtained from the Arabian Gulf was employed as the mixing water. The physical and mechanical properties of surrounding concrete are shown in Table 3. Based on its electrical conductivity and rapid chloride penetration, the corrosion protection category of the surrounding concrete was classified as 'moderate' [23,24].

3. Test plan

3.1. Sustained loading system

The loading systems were made of load-magnifying, self-reacting steel frames, as shown in Fig. 2. Similar loading frames were used by other researchers [6]. Prior to installation into the system, steel end grips were fastened on to the concrete-encased GFRP specimens. The end grips consisted of a steel pipe, 450 mm long, with inner and outer diameters of 25 and 34 mm, respectively. The pipe was carefully cleaned, threaded on the inside, and filled with epoxy to achieve adequate bond with the GFRP bar. Steel hooks were then installed on to the end grips as shown in Fig. 3. The test regions of the so-produced specimens, encased with moist seawater-contaminated concrete, were surrounded by temperature-controlled water tanks and fastened into the steel frames from the hooked ends. It should be noted that the water pressure may have an effect on the water transform performance in unsaturated concrete. Nevertheless, the concrete-encased GFRP specimens of the current study were continuously submerged during conditioning. This was done in an effort to maintain the seawater-contaminated concrete surrounding the GFRP bars saturated during conditioning. Since all specimens were surrounded by the same submerged concrete mixture during conditioning, the effect of other test variables, such as time of conditioning, temperature, and presence of sustained loading, on the durability of the

GFRP can be elucidated. More research is, however, needed to study the effect of water pressure on the durability performance of GFRP bars embedded in unsaturated concrete.

Each frame applied a sustained load of 25% of UTS to three replicate concrete-encased GFRP specimens. A total of 54 specimens (27 from Type I and 27 from Type II) were conditioned under a sustained load for 5, 10, and 15 months at conditioning temperatures of 20, 40 and 60 °C. Conditioning temperatures were selected to simulate actual year-round environmental conditions in Dubai, United Arab Emirates. Results of specimens conditioned under sustained loading are compared to those of control unconditioned specimens kept at ambient conditions prior to testing. Of the total number of bars conditioned under sustained load, 15 bars experienced creep rupture. Samples were taken from 10 key creep-ruptured specimens for further in-depth investigations.

3.2. Moisture absorption

Composite materials are susceptible to moisture permeation through the flow of water into the microgaps between polymeric chains, interfacial microcracks at the fiber-matrix transition zone, and matrix microcracks that formed during the compounding process [25]. The moisture uptake of creep-ruptured GFRP bars due to conditioning was determined according to ASTM D570 [26].

3.3. Tensile strength

The hooks at each of the end grips were removed and the GFRP reinforcing bars were carefully extracted from the concrete prisms following conditioning under sustained load. Specimens were tested to failure under uniaxial tension in displacement control at a rate of 1.5 mm/min. The tensile strength was calculated by dividing the maximum load by the average cross-sectional area

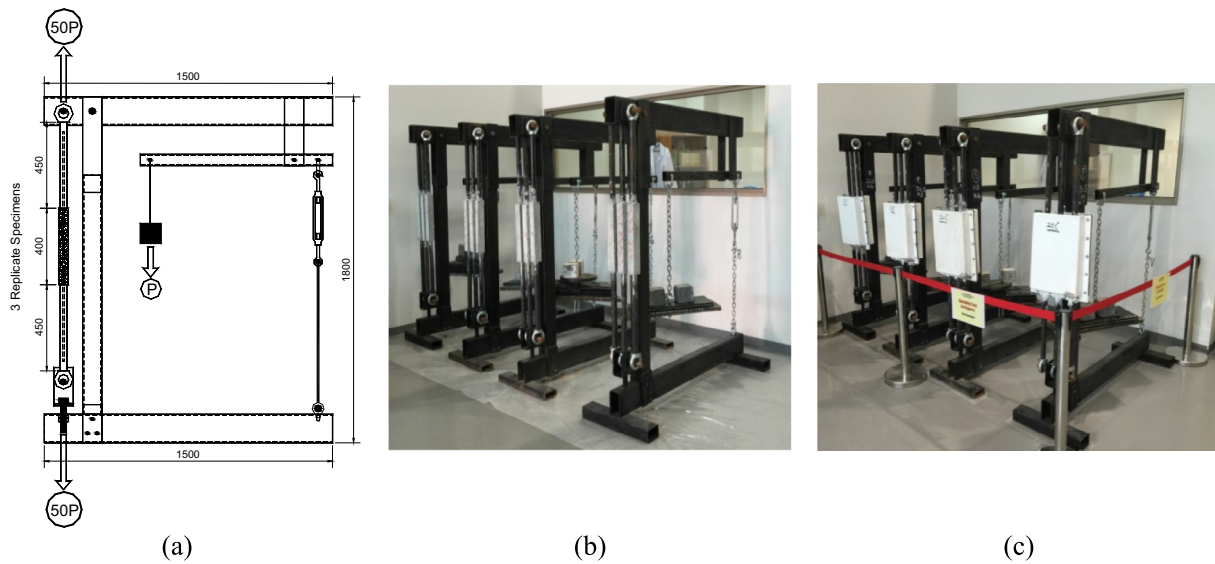


Fig. 2. Loading frame: (a) schematic, (b) specimens under load, (c) loaded specimens during conditioning.

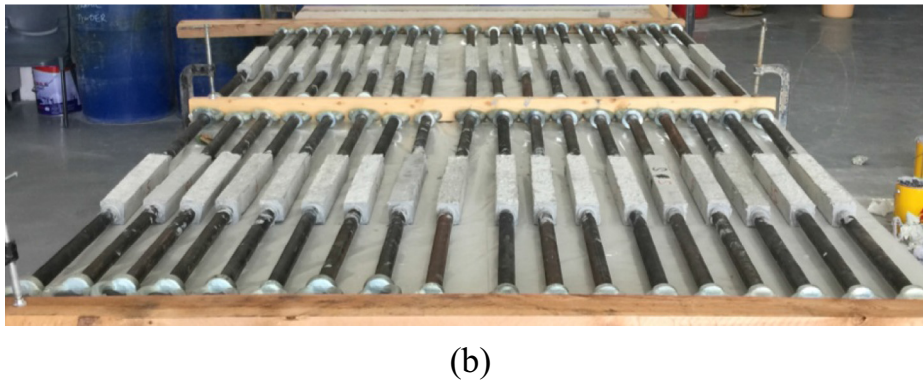
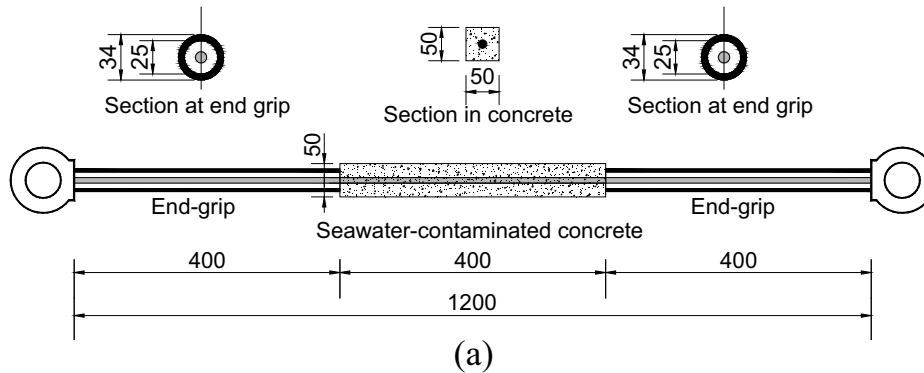


Fig. 3. Test specimens: (a) schematic, (b) after concrete casting and installation of end grips.

of the GFRP bar determined according to the test method specified by the ASTM D7205 [27].

3.4. Microstructure evaluation

Ten creep-ruptured specimens were evaluated for microstructural changes. Scanning electron microscopy (SEM) was employed to examine the changes in microstructure and morphology of control and creep-ruptured, conditioned GFRP samples by using a JEOL-JSM 6390A microscope. Specimens were taken from the

middle 100 mm, polished, and sputter coated with a thin 99.9% pure gold layer. The layer was thin enough (2–5 nm) to ensure electron conductivity during testing without covering the GFRP surface morphology. Resulting micrographs were utilized to highlight possible degradation in polymer, fiber, and fiber-matrix interface.

To measure the degradation due to hydrolysis reaction, Fourier transform infrared (FTIR) spectroscopy was conducted using a Varian 3100 FT-IR spectrometer. Specimens were taken from the middle 100 mm of the GFRP bars, pulverized, and interground with

potassium bromide (KBr) at a powder:KBr = 1:4, by mass. The mixed powder was studied over 400–4000 cm^{-1} at a resolution of 1 cm^{-1} .

Differential scanning calorimetry (DSC) was performed on powdered GFRP specimens taken from the middle 100 mm of the original samples. Specimens, of approximately 10 mg mass, were sealed in aluminum pans and prepared for analysis. Subsequently, samples were heated from 20 to 225 °C at a rate of 5 °C/min using a DSC Q2000 calorimeter equipped with a refrigerated cooling system. The glass transition temperature (T_g) was determined from experimental calorimetric curves as per ASTM E1356 [28].

4. Results and discussion

4.1. Creep rupture

GFRP specimens were conditioned under a sustained load of 25% of UTS. None of the specimens conditioned at 20 °C were creep-ruptured during conditioning. In contrast, many specimens conditioned at the higher temperatures of 40 and 60 °C were creep-ruptured prior to completing the designated conditioning duration. A summary of the number of ruptured bars is presented in Table 4. In any condition, bars failed within the part that was exposed to seawater-contaminated concrete, indicating that failure was due to conditioning-induced degradation. Two specimens from each type of GFRP bars conditioned at 40 °C were creep-ruptured within the 15-month period. Further increase in the conditioning temperature to 60 °C resulted in the rupture of 6 and 5 specimens of Type I and II GFRP, respectively, over 15 months. Evidently, increasing the temperature resulted in GFRP bars being more prone to creep-rupture under sustained loading. Type I GFRP bars conditioned at 60 °C required between 75 and 288 days (average of 195 days) to experience creep-rupture, while Type II counterparts entailed between 147 and 408 days (average of 248 days) to creep rupture.

4.2. Moisture uptake

Although GFRP reinforcing bars were concrete-encased, the concrete cracked and allowed passage of water, resulting in the uptake of moisture by the GFRP. The widths of these cracks were measured using a crackscope. Within 15 months of exposure, concrete surrounding Type I GFRP bars had an average crack width of 0.35, 0.44, and 0.50 mm when conditioned at 20, 40, and 60 °C, respectively, while those of concrete associated with Type II GFRP bars averaged at 0.51, 0.63, and 0.70 mm, respectively. Although

the same concrete was used, an increase in conditioning temperature accelerated rate of hydration at early age but created wider cracks in the concrete at later age. This is due to the accumulation of hydration products around hydrating particles, which delays the subsequent hydration and negatively affects the mechanical performance of the concrete [29].

Table 5 shows the moisture uptake of the selected creep-ruptured Type I and II GFRP bars subject to sustained load and different conditioning durations and temperatures. To simplify the comparison, samples are designated by XMYC where 'X' denotes the duration at rupture in months and 'Y' represents the temperature in degrees Celsius (°C). At the same temperature, the moisture absorption increased by increasing the time of conditioning for both types of GFRP. For Type I GFRP, a comparison between 3.8M40C and 2.5M60C shows a higher moisture absorption for samples conditioned at higher temperatures rather than longer durations. This signifies that increasing conditioning temperature results in a higher moisture uptake, owing to the formation of microcracks and an accelerated diffusion rate. Similar findings were noted in Type II GFRP reinforcing bars. For the same conditioning duration of 8.6 months, the moisture uptake at 60 °C was 2.5 times that at 40 °C. Further, samples 13.6M40C and 4.9M60C had a moisture absorption of 1.56 and 2.15%, respectively. This shows that while the conditioning duration at 60 °C was 64% shorter, the corresponding uptake was 38% higher. For any conditioning procedure, Type II GFRP bars experienced higher moisture absorption than Type I counterparts. This is owed, in one part, to the wider cracks in the concrete surrounding Type II GFRP bars, and, in another part, to the higher void content in Type II specimens. As a result, the hydrolysis reaction was further progressed in Type II GFRP bars, causing matrix softening and impairing the interfacial bond at the fiber-matrix transition zone.

4.3. Tensile strength

The tensile strengths of GFRP specimens conditioned under a sustained load are shown in Fig. 4. Specimens are designated as XMYC, where X refers to the conditioning duration in months (5, 10, and 15), and Y represents the conditioning temperature in degrees Celsius (20, 40, and 60). All specimens failed by violent rupture of fibers accompanied by fiber debonding at the fiber-matrix interface. No premature failure at the end grips was recorded in any of the tested specimens. Even though some specimens experienced creep-rupture, they were still considered in the average strength of the three replicate specimens. For instance, tensile strengths of 236 and 391 MPa of Type I GFRP bars were noted after conditioning at 5 months at 40 and 60 °C, respectively.

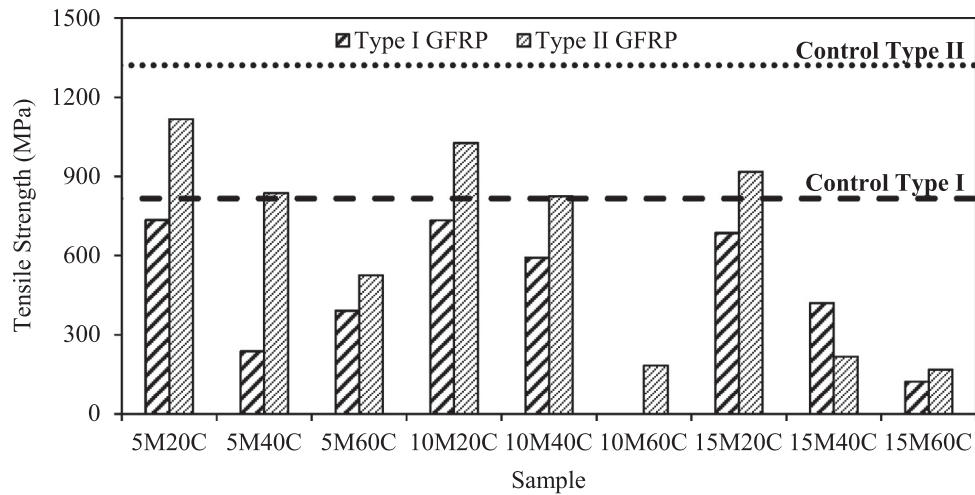
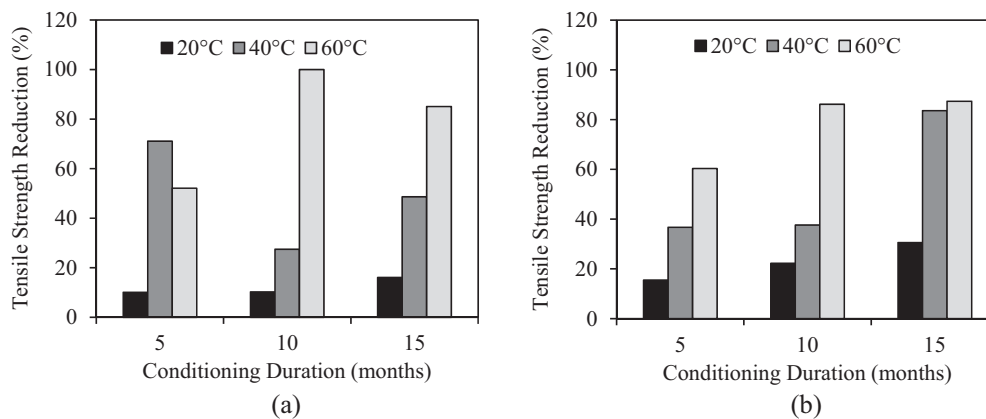
Table 4
Ruptured bars of Type I and II GFRP.

Designated Duration (months)	Temperature (°C)	GFRP Type I		GFRP Type II	
		Number of ruptured bars	Time to creep-rupture	Number of ruptured bars	Time to creep-rupture
5	20	0	–	0	–
	40	2	3.8	0	–
			3.8		–
	60	1	2.5	1	4.9
10	20	0	–	0	–
	40	0	–	0	–
	60	3	6.7	2	–
			7.9		4.9
15			8.1		4.8
	20	0	–	0	–
	40	0	–	2	8.6
			–		13.6
	60	2	9.6	2	8.6
			9.6		12.4

Table 5

Moisture uptake of GFRP bars conditioned under sustained load.

GFRP Type	Sample designation	Conditioning		Moisture absorption (%)
		Duration (months)	Temperature (°C)	
Type I	3.8M40C	3.8	40	0.54
	2.5M60C	2.5	60	1.19
	6.7M60C	6.7	60	1.29
	7.9M60C	7.9	60	1.36
	9.6M60C	9.6	60	1.95
Type II	8.6M40C	8.6	40	0.86
	13.6M40C	13.6	40	1.56
	4.9M60C	4.9	60	2.15
	8.6M60C	8.6	60	2.23
	12.4M60C	12.4	60	2.54

**Fig. 4.** Tensile strength of Type I and II GFRP bars.**Fig. 5.** Tensile strength reduction of GFRP bars conditioned under a sustained load; (a) Type I, (b) Type II.

While two bars were creep-ruptured in the former conditioning regime, only one bar was creep-ruptured in the latter. This led to a lower average tensile strength at 40 °C. Conversely, had the creep-ruptured bars been excluded from the average, the tensile strength would have been 708 and 587 MPa for respective conditioned Type I GFRP bars. Similarly, the creep-rupturing of two bars of each of Type I and II GFRP at a designated duration of 15 months of conditioning at 60 °C resulted in strengths of 122 and 167 MPa, respectively.

Fig. 5(a) shows the tensile strength reduction of Type I GFRP specimens conditioned under a sustained load. After conditioning

for 5 months, Type I GFRP bars exhibited tensile strength reductions of 10, 71, and 52% at 20, 40, and 60 °C, respectively. Samples conditioned at 40 °C suffered from most tensile strength lost, owing to the incorporation of creep-ruptured samples into the calculations. If such samples were excluded from the average tensile strength, the reduction would be 10, 16 and 27%, respectively. Extending conditioning duration led to further reduction in tensile strength. However, the effect of temperature was more pronounced due to the increase in the diffusion rate of water and to the acceleration of the hydrolysis degradation reaction. It should be also pointed out that Type I specimens conditioned at 60 °C

for 10 months experienced 100% strength loss as all three replicate specimens failed due to creep rupture.

The tensile strength reduction of Type II GFRP reinforcing bars are presented in Fig. 5(b). Specimens subjected to conditioning at 20 °C for 5, 10, and 15 months lost 15, 22, and 31% of initial tensile strength. Further reduction was noted with elevated temperatures. Type II GFRP specimens conditioned for 5 months at 40 °C exhibited higher strength reduction than those conditioned for 10 months at 20 °C indicating that strength reduction was more pronounced at higher temperatures rather than longer durations. This is indicative of the progression of hydrolysis at higher temperatures, which would result in the degradation of chemical bonds at the fiber-matrix interface.

A comparison between both types of GFRP indicates that while Type II bars presented higher initial strength, the reduction due to conditioning was more severe and thus, experienced inferior durability performance than that of their Type I counterparts. From Fig. 5(a) and (b), it can be noted that conditioning at 40 °C for 5 months and 60 °C for 10 months resulted in the creep-rupture of 2 and 3 of Type I GFRP specimens, respectively, compared to 0 and 2 for Type II GFRP at similar conditioning regimes. The corresponding tensile strength reduction of Type I seems to be higher. However, without accounting for the creep-ruptured bars, the reduction of Type I GFRP was smaller than that of Type II counterparts. Such tensile strength reduction could be caused by hydrolysis and lead to matrix softening and/or debonding at the fiber-matrix interface. Physical characterization of GFRP bars showed that Type II GFRP had higher void content and moisture absorption than those of Type I. With higher moisture uptake, Type II GFRP bars increasingly suffered from hydrolysis reaction (see FTIR analysis) with a weaker bond at the fiber-matrix interface (see SEM analysis). This phenomenon could explain the inferior durability performance of Type II GFRP bars. It is also worth noting that the difference in the diameter of both GFRP bar types used in the present study (approximately 10%) was too small to have an effect on the rate of degradation. A recent study by Benmokrane et al. [30] showed that increasing the bar diameter generally did not affect the mechanical performance of GFRP bars.

4.4. SEM analysis

Samples taken from the middle 100 mm of creep-ruptured specimens were examined by SEM. Figs. 6 and 7 represent the respective longitudinal and cross-sectional micrographs of conditioned GFRP Type I. A separation between the fiber and matrix after 3.8 months of conditioning at 40 °C was observed as shown in Fig. 6(a). The associated microgap was on the range of 2 μm , owing to the propagation of cracks from the matrix to the interface rather than directly into the fiber [31]. While Type I GFRP bars conditioned at higher temperatures [Fig. 6(b)] displayed wider gaps, the matrix showed limited disintegration. Fig. 6(c) and (d) show that increasing the time of conditioning at an elevated temperature of 60 °C further weakened the bond at the fiber-matrix interface; however, the fiber, itself, remained unaffected. This is an indication of matrix disintegration due to the hydrolysis reaction which is accelerated in hot environment, thus leading to a non-uniform distribution of loads among the fibers [1,31]. As a result, the resulting tensile strength retention was lower. Cross-sectional micrographs of Fig. 7 show signs of circumferential debonding at the fiber-matrix interface particularly at a higher temperature of 60 °C.

Fig. 8 represents the longitudinal micrographs of Type II GFRP bars conditioned at 40 °C and 60 °C. Partial matrix disintegration, gap formation and debonding at the fiber-matrix interface were observed in specimens conditioned at 40 °C [Fig. 8(a) and (b)]. These findings demonstrate that accelerated aging weakens the bond at the fiber-matrix interface. While lower temperatures could initiate separation between fiber and matrix, an increase in conditioning temperature to 60 °C widened the microgap to a few microns as shown in Fig. 8(d) and (e). Similar findings have been reported where conditioning at temperatures above 50 °C weakened the interfacial fiber-matrix bond due to transversal expansion of the matrix [32–34]. It should, however, be pointed out that the fiber was unaffected by the conditioning. The cross-sectional micrographs of Type I and II GFRP bars are presented in Fig. 9. Although conditioning-induced separation at the fiber-matrix interface was noted in all samples, it was intensified at higher conditioning temperatures. This indicates the sensitivity of the

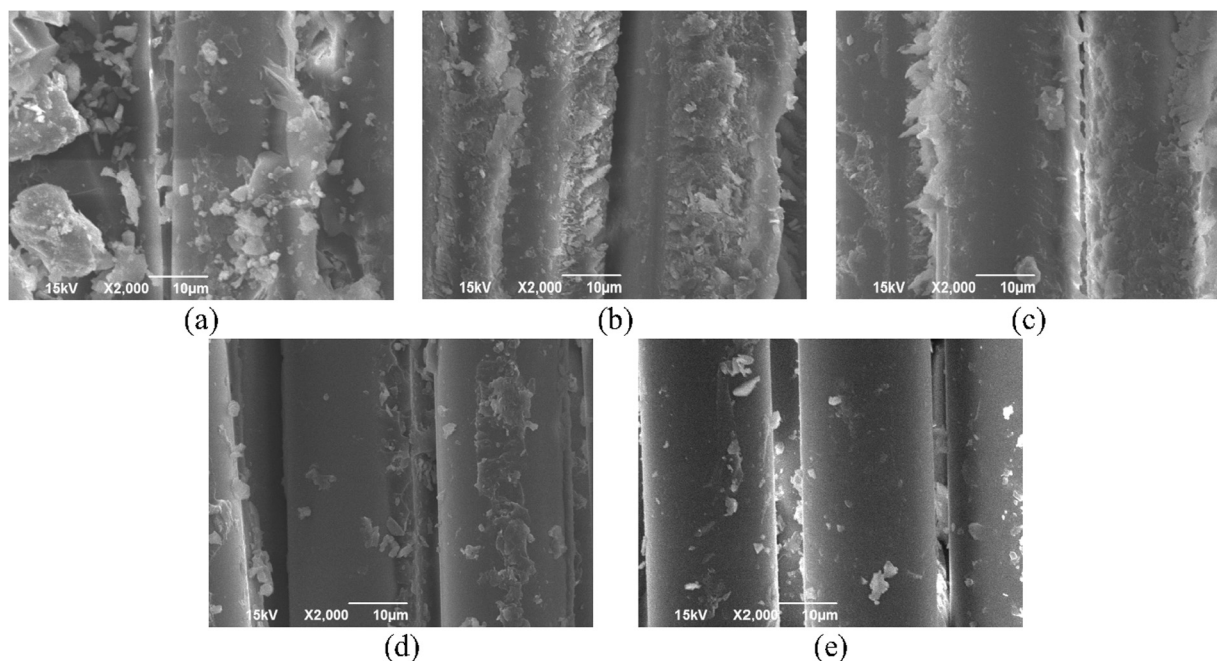


Fig. 6. Longitudinal micrographs of Type I GFRP conditioned under sustained load: (a) 3.8M40C, (b) 2.5M60C, (c) 6.7M60C, (d) 7.9M60C, (e) 9.6M60C.

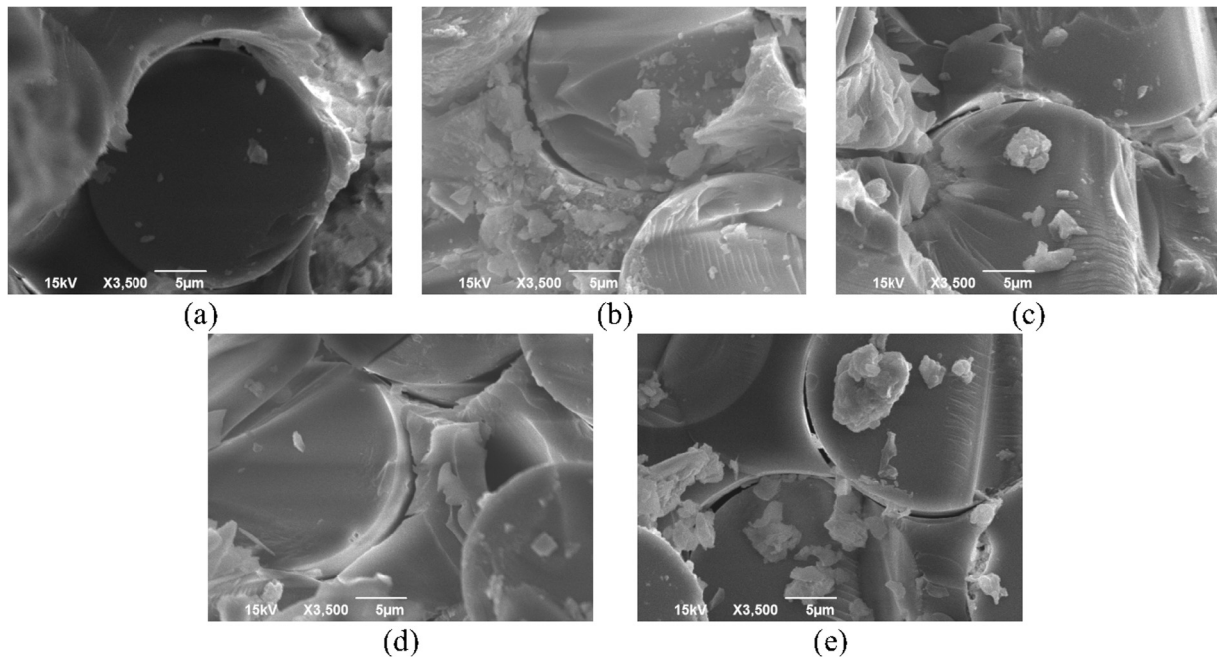


Fig. 7. Cross-sectional micrographs of Type I GFRP conditioned under sustained load: (a) 3.8M40C, (b) 2.5M60C, (c) 6.7M60C, (d) 7.9M60C, (e) 9.6M60C.

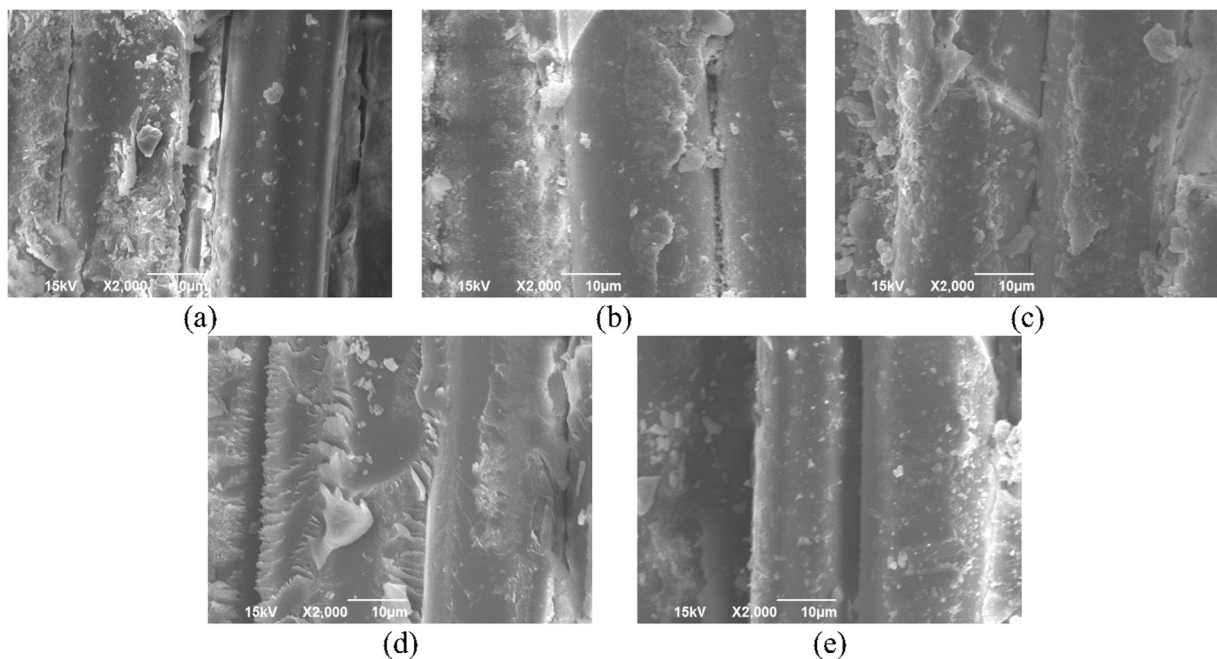


Fig. 8. Longitudinal micrographs of Type II GFRP conditioned under sustained load: (a) 8.6M40C, (b) 13.6M40C, (c) 4.9M60C, (d) 8.6M60C, (e) 12.4M60C.

fiber-matrix interface to degradation at higher conditioning temperatures rather than longer duration.

4.5. FTIR analysis

FTIR spectroscopy was employed to assess the degree of hydrolysis reaction. Mainly, two regions of the spectra were examined: 2800–3000 cm^{-1} , denoting carbon-hydrogen (CH) groups, and 3200–3600 cm^{-1} , denoting hydroxyl (OH) groups. Hydrolysis reaction results in an increase in the infrared band of OH groups, but does not affect that of CH groups. To characterize the relative quantity of hydroxyl groups in samples, the ratio of maximum peaks in

each of OH and CH band is required [1,17,35,36]. Therefore, higher OH/CH was indicative of matrix degradation.

Table 6 shows the band ratios of conditioned and control samples. Clearly, hydrolysis reaction has progressed as an increase in temperature and/or duration has resulted in an increase in the OH/CH ratio. After conditioning for 3.8 months at 40 °C, Type I GFRP showed approximately a 4.5% increase in OH/CH from a value of 1.07–1.12. Samples conditioned at 60 °C, but for 2.5 months, experienced further development of OH groups with a OH/CH ratio of 1.22. As for Type II GFRP bars, an increase in conditioning duration from 8.6 to 13.6 months at 40 °C led to an increase in OH/CH from 1.14 to 1.19. The effect of temperature however was more

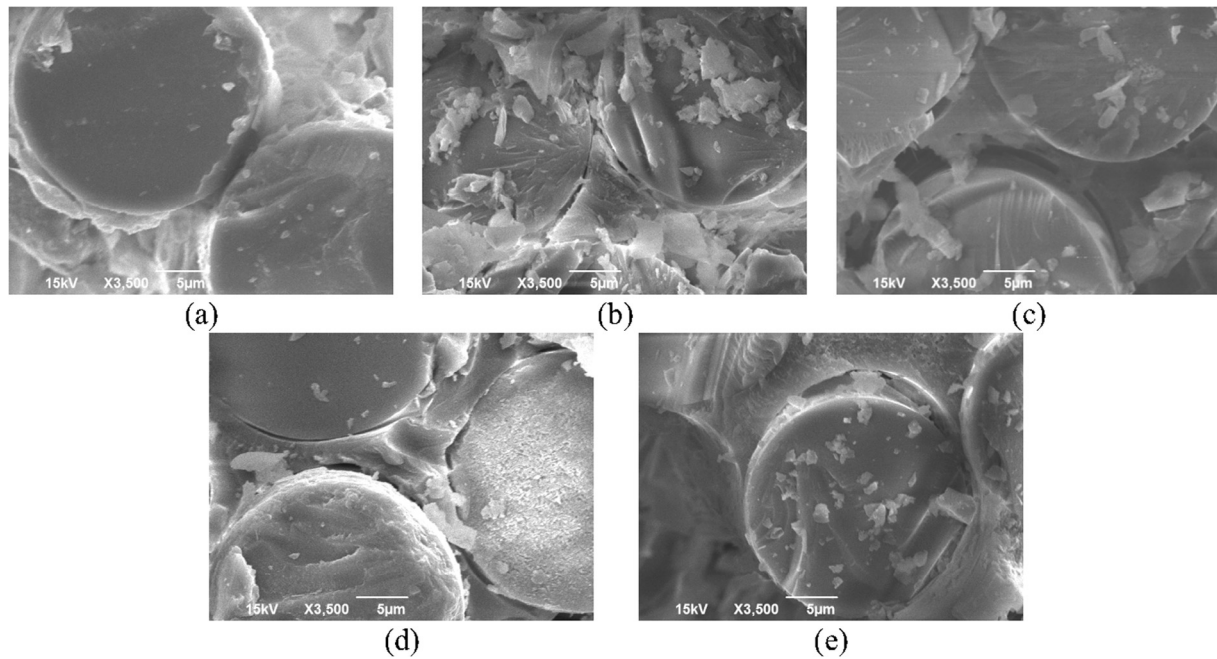


Fig. 9. Cross-sectional micrographs of Type II GFRP conditioned under sustained load: (a) 8.6M40C, (b) 13.6M40C, (c) 4.9M60C, (d) 8.6M60C, (e) 12.4M60C.

Table 6

Band ratios of conditioned and control samples.

GFRP Type	Conditioning		Peaks		OH/CH	Increase (%) ^a
	Duration (months)	Temp. (°C)	CH	OH		
Type I	Control	Ambient	3.62	3.88	1.07	–
	3.8	40	4.47	5.00	1.12	4.36
	2.5	60	3.13	3.83	1.22	14.16
	6.7	60	4.84	6.02	1.24	16.05
	7.9	60	4.92	6.20	1.26	17.57
	9.6	60	6.56	8.33	1.27	18.47
Type II	Control	Ambient	2.77	1.62	0.59	–
	8.6	40	4.30	4.91	1.14	94.78
	13.6	40	4.12	4.89	1.19	102.69
	4.9	60	4.49	5.14	1.14	95.18
	8.6	60	5.21	6.00	1.15	96.53
	12.4	60	5.58	7.11	1.27	117.44

Note: CH = carbon-hydrogen groups; OH = hydroxyl groups.

^a Change in OH/CH ratio of conditioned specimens with respect to that of the control specimen.

apparent as samples conditioned at 60 °C for 4.9 months resulted in a similar OH/CH as those conditioned at 40 °C for 8.6 months. This signifies that conditioning temperature was more effective at advancing the hydrolysis reaction compared to conditioning duration. Similar findings have been reported in other work [1,37].

Table 6 also presents the percent increase in the ratio of OH/CH of creep-ruptured conditioned GFRP specimens relative to the control unconditioned samples. Conditioned Type II GFRP bars developed more hydroxyl groups than Type I counterparts. While Type I GFRP samples conditioned at 60 °C for 9.6 months reported an increase of 18%, Type II GFRP bars could reach 95% increase within 4.9 months at the same conditioning temperature. It seems that the higher void ratio of Type II bars combined with the wider cracks in the surrounding concrete resulted in a higher moisture uptake, progression of hydrolysis, and further development of hydroxyl groups. This, in turn, led to higher tensile strength reduction.

4.6. DSC analysis

The glass transition temperature (T_g) of control and creep-ruptured conditioned samples of Type I and II GFRP bars are summarized in Table 7. Two scans were performed for each sample: the first scan was to determine the difference between the T_g of control and that of conditioned samples as the decrease in T_g is indicative of plasticization, while the second scan was to investigate the degradation mechanism. After 3.8 months at 40 °C, Type I GFRP samples recorded a T_g of 95 °C. This represents a 6 °C drop compared to the unconditioned control. Conditioning at higher temperatures resulted in a further decrease in T_g down to 90 °C. Similarly, Type II GFRP bars showed a decline in T_g from 125 °C to 90 °C after conditioning for 12.4 months at 60 °C. Such decrease in T_g was indicative of matrix plasticization. Although a T_g of 90 °C was recorded for both types of conditioned GFRP samples, the drop in Type II was 35 °C compared to 11 °C in Type I counterparts. This

Table 7

Glass transition temperature of GFRP bars using DSC analysis.

GFRP Type	Conditioning		T_g (°C)	
	Duration (months)	Temp. (°C)	1st Scan	2nd Scan
Type I	Control	Ambient	101	106
	3.8	40	95	104
	2.5	60	94	105
	6.7	60	91	104
	7.9	60	90	105
	9.6	60	90	105
Type II	Control	Ambient	125	125
	8.6	40	98	125
	13.6	40	93	126
	4.9	60	95	125
	8.6	60	92	126
	12.4	60	90	125

Table 8

Effect of sustained load on tensile strength of conditioned bars.

GFRP type	Conditioning		Strength retention (%)		Ratio SR_1/SR_u^a (SR_2/SR_u^b)	Number of ruptured bars
	Duration (months)	Temp. (°C)	Sustained load	No load		
Type I	5	20	90	98	0.92	–
		40	28 (84) ^c	92	0.32 (0.91)	2
		60	48 (72) ^c	90	0.53 (0.80)	1
	10	20	90	98	0.91	–
		40	72	95	0.76	–
		60	0	94	0.00	3
	15	20	84	98	0.86	–
		40	51	92	0.55	–
		60	15 (45) ^c	85	0.18 (0.53)	2
Type II	5	20	84	84	1.00	–
		40	63	77	0.82	–
		60	39 (58) ^c	73	0.63 (0.80)	1
	10	20	78	79	0.99	–
		40	62	66	0.94	–
		60	14 (42) ^c	50	0.28 (0.84)	2
	15	20	70	73	0.95	–
		40	16 (48) ^c	68	0.24 (0.71)	2
		60	12 (36) ^c	53	0.42 (0.68)	2

^a SR_1/SR_u : Ratio of strength retention of loaded specimens to that of unloaded specimens.^b SR_2/SR_u : Ratio of strength retention of non-ruptured, loaded specimens to that of unloaded specimens.^c Strength retention of non-ruptured, loaded specimens.

provides evidence of the inferior durability performance of Type II GFRP bars, which is also well-aligned with FTIR results. As voids formed in the matrix microstructure (see *SEM analysis*), debonding at the fiber-matrix interface took place. The polymer matrix was thus exposed to more water/moisture, resulting in higher water absorption. Consequently, the polymer structure was modified and the T_g was reduced.

Samples were re-heated during the second scan. As water evaporated, it reversed the plasticizing effect, and thus, T_g values of conditioned and unconditioned samples were similar. Results of Table 7 show that the T_g was unaffected by conditioning during the second scan. This implies that the polymer matrix was not subjected to an irreversible chemical degradation. It should however be noted that the T_g of Type I GFRP control was higher in the second scan due to a post-curing mechanism. Other researchers reported similar findings [6].

4.7. Comparison with unloaded conditioned specimens

Based on previous mechanical performance and microstructure investigation of Type I and II GFRP reinforcing bars conditioned without a sustained load [38], results of specimens of the current study conditioned under a sustained load are compared to those

of their counterparts conditioned without load. As shown in Table 8, Type I and II GFRP specimens conditioned without a sustained load retained up to 85 and 50% of their initial tensile strength after conditioning, respectively. The presence of a sustained load of 25% of UTS during conditioning decreased the tensile strength retention. For instance, Type I GFRP samples conditioned at 40 °C for 10 months experienced respective tensile strength retentions of 72 and 95% with and without a sustained load. The corresponding strength retention ratio of loaded to unloaded specimens, designated by SR_1/SR_u , was 0.76. While conditioning without a load was detrimental to GFRP bars, the presence of a sustained load during conditioning promoted degradation and loss in tensile strength.

The ratio SR_1/SR_u considers ruptured GFRP bars in the calculations of tensile strength retention. This causes a significant decrease in tensile strength retentions of the specimens conditioned under a sustained load. To investigate the effect of sustained loading on the tensile strength retention, another ratio, SR_2/SR_u , was calculated. This ratio represents the tensile strength retention of non-ruptured GFRP bars conditioned under a sustained load to that of GFRP bars conditioned without a load. As presented in Table 8, even though the strength retentions of Type I GFRP bars were higher than those of Type II equivalents, the SR_2/SR_u ratio

was generally lower. Hence, it could be concluded that Type I GFRP was affected more by the presence of a sustained load. The strength retention for Type I and II GFRP conditioned with and without a load decreased while increasing conditioning temperature and duration. It is clear that the ratio (SR_2/SR_1) also decreases with an increase in the conditioning temperature. This implies that the degradation was intensified when a sustained load was applied at elevated conditioning temperatures.

The moisture uptakes of GFRP specimens conditioned under a sustained load were higher than those of their unloaded counterparts reported in [38]. For instance, the moisture absorption of Type I GFRP bars conditioned under a sustained load for approximately 10 months at 60 °C (9.6M60C) was 1.95% (Table 5). El-Hassan et al. [38] reported that Type I GFRP bars conditioned without load experienced a moisture uptake of only 1.17%. Similarly, Type II GFRP bars conditioned under a sustained load after approximately 5 months of exposure at 60 °C (4.9M60C) exhibited a moisture absorption of 2.15% (Table 5). Their counterparts conditioned without load exhibited a moisture absorption of 0.75% only [38]. This indicates that the presence of a sustained load during conditioning increased the moisture absorption. Although the same surrounding concrete was used, loading the GFRP bars initiated cracks into the concrete and further exposed the bars to water/moisture, thus increasing the absorption. Accordingly, the degree of hydrolysis reaction was higher, with further development of hydroxyl groups and lower T_g , as evidenced by FTIR and DSC analysis, respectively.

The ACI 440.1R [16] limits the level of sustained load to 14% of UTS, while accounting for environmental conditioning. Additionally, the environmental reduction factor recommended by the ACI 440.1R [16], 0.7, indicates that the retained tensile strength should be greater than 70% of the initial strength. In this work, a sustained load of 25% of UTS was taken to investigate the possible increase in the allowable stress. At 20 °C, Type I and II GFRP bars retained 84 and 70% of the initial tensile strength after 15 months of conditioning under 25% sustained load. At 40 and 60 °C, respective Type I GFRP tensile strength retentions (non-ruptured bars only) were 51 and 45%, while those of Type II GFRP were 48 and 36%, respectively. These findings show that GFRP reinforcing bars exposed to a temperature of 20 °C can be stressed to up to 25% of UTS without compromising their durability. However, the limit set by the ACI 440.1R [16] cannot be increased for GFRP bars subject to elevated conditioning temperatures of 40 and 60 °C.

4.8. Long-term durability prediction

The accelerated aging test data along with the Arrhenius concept were employed to develop a durability design model that

can predict the tensile strength retention of both types of GFRP bars in moist seawater-contaminated concrete. Arrhenius concept (Eq. (1)) assumes that while a time and temperature-independent degradation mechanism dominates during conditioning, its rate accelerates at higher temperatures [39]. The natural logarithm ($\ln$) of the time needed to reach a particular tensile strength retention versus the inverse of temperature (i.e. Arrhenius-type relationship), in terms of 1000/K, was employed to predict the service life of Type I and II GFRP bars at the average annual temperature in Dubai of 27 °C.

$$\ln \frac{1}{k} = \frac{E_a}{RT} - \ln(A) \quad (1)$$

Using the Arrhenius plots for actual year-round environmental conditioning in Dubai, i.e. 20, 40, and 60 °C, the service life of GFRP bars can be extrapolated for any temperature. Predictions were thus made for tensile strength retention over immersion time at 27 °C and plotted in Fig. 10(a). It can be seen that the tensile strength retention of Type I GFRP bars in moist seawater-contaminated concrete without load dropped to 80% after about 10 years. Conditioning these bars under a sustained load of 25% of UTS resulted in a 50% decrease in tensile strength in approximately 3 years. This demonstrates the negative effect of sustained loading on the service life of Type I GFRP bars. Fig. 10(b) presents the change in tensile strength of Type II GFRP bars over exposure time. The effect of sustained load on the service life of Type II GFRP bars was less significant compared to that of Type I counterparts. The tensile strength retention of Type II GFRP bars would drop to 50% in about 1.5 and 2.5 years when exposed to moist seawater-contaminated concrete with and without a sustained load, respectively. It is possible that conditioning without load caused significant degradation in the tensile strength of Type II GFRP due to a deterioration of the chemical bonds at the fiber-matrix interface. As a result, the polymer structure was severely weakened to the extent that the presence of a sustained load did not further progress the degradation. With much lower tensile strength retention, Type II GFRP bars presented much inferior durability performance compared to Type I GFRP counterparts, even though as-received mechanical properties were superior. Nevertheless, it is worth noting that GFRP bars presented in this study were encased in seawater-contaminated concrete and continuously immersed in tap water during conditioning. This is not the normal practical setting; instead, concrete is usually exposed to different levels of relative humidity. Furthermore, reinforcing bars in real structures would have the full depth of the structural element from the internal side of the element, which would reduce the exposure severity. As a result, the predictions presented herein are considered conservative.

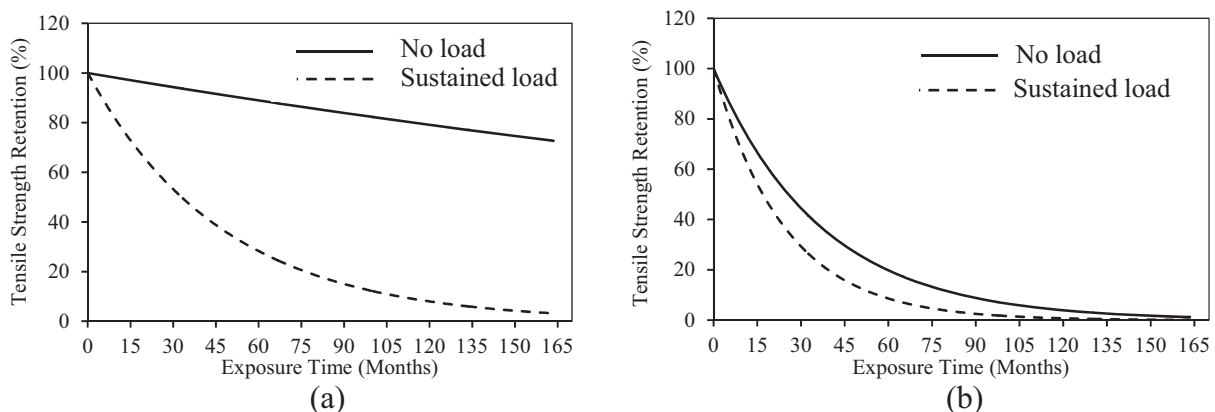


Fig. 10. Durability design models in moist seawater-contaminated concrete located in Dubai: (a) Type I GFRP, (b) Type II GFRP bars.

5. Conclusions

Tensile strength retentions and microstructural characteristics of two types of commercially-produced GFRP bars conditioned in moist seawater-contaminated concrete under a sustained load of 25% of their ultimate tensile strength were evaluated in this paper. In real structures, tensile forces are typically resisted by the reinforcing bars. The effect of presence of a sustained tensile stress during conditioning is investigated in the current study to better represent field conditions. Results of specimens of the current study conditioned under a sustained load were compared to those reported by El-Hassan et al. [38] for counterpart specimens conditioned without load. Accelerated aging test data were employed to develop durability design models of both types of GFRP in moist seawater-contaminated concrete. Main conclusions of the work are:

1. None of the specimens conditioned at 20 °C under a sustained load were creep-ruptured during conditioning. Increasing the conditioning temperature resulted in GFRP bars being more prone to creep-rupture under a sustained load. Two specimens from each type of GFRP bars conditioned at 40 °C were creep-ruptured within a 15-month period. Further increase in the conditioning temperature to 60 °C resulted in the rupture of 6 and 5 specimens of Type I and II GFRP, respectively, over 15 months.
2. At the same temperature, the moisture absorption increased by increasing the time of conditioning for both types of GFRP. Increasing the conditioning temperature resulted in a higher moisture uptake. Type II GFRP bars experienced higher moisture absorption than that of their Type I counterparts due to formation of wider cracks in the surrounding concrete in addition to their higher void content. The moisture uptakes of GFRP specimens conditioned under a sustained load were higher than those of their unloaded counterparts.
3. At a conditioning temperature of 20 °C, Type I and II GFRP bars retained 84 and 70% of their initial tensile strength after 15 months of conditioning under a sustained load. At the higher temperatures of 40 and 60 °C, many specimens were creep-ruptured prior to completing the designated conditioning duration, which resulted in significant tensile strength reductions.
4. At a conditioning temperature of 60 °C, trivial tensile strength retentions of 15 and 12% were recorded for Type I and Type II GFRP bars, respectively, after 15 months of conditioning under a sustained load. When results of the creep-ruptured bars were excluded from the average value of the three replicate specimens, respective tensile strength retentions of 45 and 36% were recorded for Type I and Type II GFRP bars, respectively. The lower tensile strength retentions recorded for Type II GFRP bars were consistent with their higher moisture absorption values relative to those exhibited by their Type I counterparts.
5. Based on a comparative analysis between results of loaded and unloaded specimens, it was found that while conditioning without a load was detrimental to GFRP bars, the presence of a sustained load during conditioning intensified the loss in the tensile strength. The additional degradation caused by the presence of a sustained load during conditioning was aggravated at the higher temperatures.
6. Using Arrhenius concept for actual year-round environmental conditioning in Dubai, it was found that the tensile strength retention of Type I GFRP bars in moist seawater-contaminated concrete without load would drop to 80% after about 10 years. Type I GFRP bars would lose 50% of the original tensile strength in approximately 3 years when subjected to moist seawater-contaminated concrete under a sustained load of 25%. The tensile strength retention of Type II GFRP bars would drop to 50% in

about 1.5 and 2.5 years when exposed to moist seawater-contaminated concrete with and without a sustained load, respectively. Results of the long-term prediction reported in the current study are considered conservative because the concrete-encased GFRP specimens were continuously immersed in water during conditioning while in practical settings, concrete is usually exposed to different levels of relative humidity.

7. While the ACI 440.1R [16] recommends a maximum sustained load value of 14%, current findings advocate for a possible increase of this limit to 25% for regions with average temperatures below 20 °C because the tensile strength retention for all specimens conditioned at 20 °C was $\geq 70\%$. Nevertheless, for hot regions with annual mean temperatures above 20 °C, results of the current study show that GFRP bars are not durable enough to withstand a sustained load of 25%.

Results of the present study provided insight into the durability performance of GFRP bars conditioned in moist seawater-contaminated concrete under sustained loading. Conditioning of GFRP bars in such environmental conditions is believed to be more representative of the real life situation than immersing the bars in seawater or an alkaline solution. Further research is still needed to study the effect of varying the thickness of the concrete cover on the durability of GFRP bars in severe environments. The effects of the relative humidity and water pressure on the durability performance of GFRP bars embedded in unsaturated concrete should also be investigated in future studies.

Conflict of Interest

The authors wish to confirm that there are no known conflicts of interest associated with this publication.

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